

USED CARS PRICE ANALYSIS

DATA CLEANING STEPS

IMPORTING THE REQUIRED LIBRARIES

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
import glob
```

CLEANING

STEP 1 : LOADING THE DATA SET

- `df=pd.read_csv(r"E:\Used_Cars_Analysis\data\raw\cars24_agra_raw.csv")`

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LOAD ALL THE CITIES DATASETS

STEP 2 : DATA UNDERSTANDING / DATA EXPLORATION (EDA)

- `df.head()` → First 5 rows
- `df.tail()` → Bottom 5 rows
- `df.info()` → Summary of the DataSet
- `df.describe(include="all")` → Statistical Summary of the DataSet
- `df.shape` → Size of the DataSet (Rows & Columns)

STEP 3 : DATA CLEANING / DATA WRANGLING

- Checking Null Values
`df.isnull().sum()`
- Checking Duplicates
`df.duplicated().sum()`
- Column names
`df.columns`
- Dropping Inessential Columns
`df.drop('Unnamed: 0',axis=1,inplace=True)`
- Renaming Column Names
`df.rename(columns={'Distance Driven':'KMS_Driven'},inplace=True)`
- Standardizing the Column Names
`df.columns = (df.columns.str.lower().str.replace(" ", "_"))`
- Standardizing the Data
`df['kms_driven'] = (df['kms_driven'].str.replace(" km", "").str.replace(", ", "").astype(int))`
`df['emi']=(df['emi'].str.replace("EMI ₹", "").str.replace(", ", "").str.replace("/m*", "").astype(int))`
`df['price_num'] = df['price'].str.extract(r'([\d.]+)').astype(float)`
`df['is_crore'] = df['price'].str.contains(' Crore', case=False, regex=True)`
`df['price_lakhs'] = df['price_num']`
`df.loc[df['is_crore'], 'price_lakhs'] = df['price_num'] * 100`
`df.drop(['price', 'price_num', 'is_crore'],axis=1,inplace=True)`
`df.rename(columns={'price_lakhs':'price'},inplace=True)`
`df['price']=(df['price']*100000)`

STEP 4 : FEATURE ENGINEERING

- Declaring Current Year
`current_year=2026`
- Creating car_age_years Column
`df['car_age_years']=current_year - df['year_of_manufacture']`
- Creating city column
`df['city'] = ' city_name '`

STEP 5 : OUTLIERS DETECTION

- `num_cols = ['kms_driven','emi','price']`
`for col in num_cols:`
 `Q1 = df[col].quantile(0.25)`
 `Q3 = df[col].quantile(0.75)`
 `IQR = Q3-Q1`
 `lower = Q1-1.5*IQR`
 `upper = Q3+1.5*IQR`
 `outliers = df[(df[col]<= lower) | (df[col]>= upper)]`
`outliers`

#As all outliers present in the data are real and normal , they are not errors so we are not dropping them .

STEP 6 : FINAL VALIDATION

- `df.isnull().sum()`
- `df.duplicated().sum()`

STEP 7 : SAVING THE CLEANED FILE

- `df.to_csv("cars24_cityname_cleaned.csv")`

CONTINUE SAME STEPS FOR ALL THE CITIES DATA

APPENDING ALL THE CITIES

STEP 1 : IMPORTING THE REQUIRED LIBRARY

- `import glob`

STEP 2 : APPENDING

- `all_data =
pd.concat(map(pd.read_csv,glob.glob(r"E:\Used_Cars_Analysis\data\cleaned/*.csv"))
,ignore_index=True)`

STEP 3 : SAVING THE FILE

- `all_data.to_csv("cars_24_all_cities.csv")`

STEP 4 : BASIC EXPLORATION

- `all_data.head()` → First 5 rows
- `all_data.tail()` → Bottom 5 rows
- `all_data.info()` → Summary of the DataSet
- `all_data.describe(include="all")` → Statistical Summary of the DataSet
- `all_data.shape` → Size of the DataSet (Rows & Columns)